

FAYETTE REGIONAL, TX, USA

WMO: 722554

Lat: 29.908N

Lon: 96.950W

Elev: 99

StdP: 100.14

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 12978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.3	-11.1	1.5	9.0	-7.5	2.0	8.9	9.8	14.7	8.8	13.4	1.6	0	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	37.6	24.0	36.7	24.2	35.5	24.5	27.2	31.3	26.8	31.2	26.4	31.1	3.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.0	28.2	25.8	21.4	28.1	25.3	20.7	27.9	86.0	31.5	84.1	31.2	82.5	30.9	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.4	-3.8	39.4	2.2	1.6	-5.4	40.5	-6.7	41.4	-7.9	42.3	-9.5	43.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-4.7	27.8	2.3	1.2	-6.4	28.6	-7.7	29.3	-9.0	30.0	-10.6	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.1	11.4	13.4	17.4	20.8	24.8	28.2	29.2	29.9	27.0	22.0	16.6	12.3
	DBStd	7.63	5.19	5.66	4.74	3.78	3.09	1.82	1.63	1.73	2.49	4.30	5.04	5.21
	HDD10.0	119	44	28	5	1	0	0	0	0	0	7	34	
	HDD18.3	778	223	154	75	19	2	0	0	0	17	90	198	
	CDD10.0	4183	88	124	234	325	460	547	595	616	511	373	206	105
	CDD18.3	1800	8	16	46	94	203	297	337	357	261	131	39	11
	CDH23.3	19154	32	79	239	697	1914	3420	4057	4502	2689	1209	259	56
CDH26.7	8947	1	10	33	171	734	1688	2106	2476	1257	423	44	6	
Wind	WSAvg	2.4	2.6	2.9	2.9	3.0	2.8	2.3	2.0	1.9	1.6	1.9	2.3	2.5
Precipitation	PrecAvg	992	81	69	70	82	99	106	64	62	90	121	93	79
	PrecMax	1384	223	203	183	231	279	266	184	161	193	342	304	245
	PrecMin	545	2	4	8	9	4	1	3	8	8	0	9	24
	PrecStd	265	53	49	41	58	72	66	50	46	60	102	79	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	28.0	29.1	32.3	34.9	38.0	38.2	39.2	37.5	33.9	29.8	27.3
		MCWB	18.4	19.6	20.7	21.6	23.7	24.0	24.1	24.4	22.3	23.6	22.0	20.3
	2%	DB	23.9	25.9	27.5	30.1	33.0	36.3	37.1	37.9	35.9	32.3	27.8	24.9
		MCWB	17.0	18.7	20.0	21.6	23.7	24.0	24.1	24.5	23.5	23.6	21.0	18.8
	5%	DB	22.3	23.8	26.2	28.6	32.2	34.9	36.0	37.1	34.0	31.0	26.3	22.9
		MCWB	17.6	18.4	19.5	21.5	23.6	24.6	24.6	24.4	24.1	22.9	20.1	18.5
	10%	DB	20.6	22.2	24.3	27.3	31.0	33.6	34.9	36.0	32.7	29.1	24.8	21.7
		MCWB	16.4	18.4	19.2	20.4	23.4	24.9	25.0	24.5	23.7	21.7	19.7	17.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	22.2	23.0	25.1	26.6	27.9	27.5	27.5	27.3	26.6	24.5	22.8
		MCDB	23.3	24.8	26.3	28.5	31.0	31.5	31.5	31.5	30.3	29.1	27.3	24.8
	2%	WB	20.1	21.2	22.2	23.9	25.8	27.0	26.9	26.9	26.6	25.7	22.9	21.4
		MCDB	22.6	23.7	25.0	27.3	29.9	31.3	31.4	31.5	30.3	29.0	25.5	23.8
	5%	WB	18.9	20.1	21.5	23.2	25.2	26.5	26.6	26.6	26.1	24.9	21.9	20.2
		MCDB	21.5	22.9	24.1	26.6	29.4	31.1	31.3	31.5	30.1	28.4	24.8	22.3
	10%	WB	17.4	18.9	20.6	22.4	24.6	26.0	26.2	26.1	25.6	24.1	20.9	18.5
		MCDB	19.4	21.4	23.1	25.6	28.5	30.4	31.0	31.3	29.7	27.4	23.5	20.9
Mean Daily Temperature Range	5% DB	MDBR	11.2	10.9	11.0	11.2	10.3	10.3	10.7	11.6	10.8	11.9	11.6	11.5
		MCDBR	13.1	12.2	11.2	11.7	11.3	12.3	12.6	13.1	12.5	11.8	12.0	13.0
		MCWBR	7.9	7.1	5.5	5.5	3.9	3.1	2.8	2.8	3.6	4.9	6.6	8.8
	5% WB	MCDBR	10.0	10.2	9.1	9.1	8.9	9.5	10.2	11.4	9.7	9.4	9.9	11.2
		MCWBR	7.4	7.2	6.0	5.4	3.8	3.2	3.0	3.0	3.7	5.2	6.2	9.3
Clear-Sky Solar Irradiance	taub	0.337	0.343	0.368	0.412	0.463	0.441	0.460	0.449	0.428	0.377	0.371	0.327	
	taud	2.469	2.468	2.399	2.293	2.193	2.307	2.285	2.304	2.320	2.445	2.396	2.513	
	Ebn at Noon	896	923	917	884	836	850	833	839	847	874	847	890	
	Edh at Noon	93	101	116	133	148	131	134	130	124	102	99	85	
All-Sky Solar Radiation	RadAvg	2.92	3.33	4.41	5.31	5.79	6.42	6.47	6.12	5.12	4.30	3.21	2.59	
	RadStd	0.30	0.62	0.40	0.44	0.44	0.44	0.53	0.34	0.48	0.54	0.36	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	+88

Nomenclature: See separate page